

Notes: Falck, Gold, and Heblich (AER, 2014)

E-lections: Voting Behavior and the Internet

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1 Big picture

- **Question.** How did the arrival of broadband Internet affect voting behavior? In particular, did Internet availability change voter turnout, and did it systematically help particular types of political parties?
- **Setting.** The paper studies German municipalities during the early broadband period. The main comparison is between election outcomes in the **pre-Internet period** (1995–1999) and the **broadband Internet period** (2004–2008).
- **Core idea.** The Internet lowers the cost of acquiring political information, but it also creates many new entertainment options and can crowd out older media such as television or newspapers. Therefore, the effect on political participation is theoretically ambiguous.
- **Core empirical challenge.** Broadband infrastructure was not rolled out randomly. Telecom operators had stronger incentives to expand first in municipalities with richer, younger, more educated, or otherwise more profitable residents. These characteristics are also correlated with voting behavior, so ordinary cross-sectional regressions would confound Internet effects with selection into broadband access.
- **Main contribution.** The paper uses historical and technological peculiarities of Germany's preexisting voice-telephony network as instruments for broadband Internet availability. These network features affected the feasibility of first-generation fixed-line DSL, but were determined before broadband Internet existed.
- **Key identification objects.**
 - In West Germany, DSL was difficult to provide when the copper-wire distance from a municipality to its main distribution frame exceeded about **4,200 meters**.
 - In another West German design, some municipalities above the 4,200-meter threshold could be reconnected to a closer distribution frame, while others could not.
 - In East Germany, some areas had been supplied with **OPAL** fiber technology after reunification. OPAL was initially considered advanced, but later turned out to be incompatible with DSL.
- **Why this setting is useful.** Germany had nearly universal telephone infrastructure before the Internet era. First-generation DSL used the old copper network, so accidental features of that old network generated quasi-random variation in broadband availability across otherwise comparable municipalities.
- **Main empirical question.** When DSL availability rises for technological reasons, do turnout and party vote shares change?
- **Main empirical results.**
 - DSL availability has a **negative effect on voter turnout** in West Germany.
 - The preferred West German IV estimates imply that moving from no DSL availability to full DSL availability lowers turnout by roughly **3–5 percentage points**.
 - The effect is concentrated in **nonlocal elections** – federal and state elections – rather than local elections.
 - The paper finds **no robust evidence** that broadband systematically benefits established parties, fringe parties, or any particular ideological camp.

- In East Germany, the estimates are imprecise and do not show a systematic turnout effect.
- Mechanism evidence suggests that broadband **crowds out TV consumption** and increases entertainment time, while there is little evidence that it reduces newspaper circulation.
- **Magnitude.** In the West German samples, DSL expansion can explain more than **one-third** of the observed decline in turnout between the pre-Internet and Internet periods. Among individuals exposed to broadband, the implied demobilization effect is about **8 percentage points**.
- **Economic takeaway.** A new medium can reduce political participation if its entertainment value and its crowding-out of incumbent informative media dominate its ability to lower information costs. The Internet did not simply make voters better informed; in this early period, it appears to have distracted or redirected attention away from politics.

2 Data and measurement (key objects)

2.1 Broadband Internet data

- **Source.** Broadband availability comes from the German broadband atlas, *Breitbandatlas Deutschland*, published by the Federal Ministry of Economics and Technology.
- **Geography.** The unit is the German municipality. Municipalities are the lowest level of territorial division in Germany; the average municipality corresponds roughly to a circle with radius **3.1 km**.
- **Coverage.** The broadband data cover the universe of German municipalities, using 2008 territorial boundaries, for **2005–2008**.
- **Key variable.** The treatment variable is technical DSL availability:

DSL_i = share of households in municipality i for which DSL is technically available.

- **Speed threshold.** Operators report coverage at a minimum downstream rate of **384 kb/s**.
- **Why DSL.** DSL was by far the dominant broadband technology in Germany in this period. The paper therefore focuses on fixed-line DSL rather than cable, satellite, or mobile broadband.
- **Important interpretation.** The paper estimates the effect of **DSL availability**, not directly the effect of DSL usage. Later, the SOEP survey evidence shows that the same technological instruments strongly predict actual household DSL subscriptions, so availability is a good proxy for access and use.
- **Historical timing.** Deutsche Telekom first sold DSL to private households in 1999. The period 2005–2008 is therefore an early broadband diffusion period, before social media and Web 2.0 were central in German election campaigns.

2.2 Election data

- **Source.** Election results are compiled from the statistical offices of the German states.
- **Election levels.** The paper uses three main levels of governance:

- federal elections;
- state elections;
- local elections.
- **Outcome variables.** The paper studies five election outcomes:
 - voter turnout;
 - vote share of established parties;
 - vote share of small parties;
 - vote share of right-fringe parties;
 - vote share of left-fringe parties.
- **Main periods.**
 - **Pre-Internet period:** election cycle 1995–1999.
 - **Internet period:** election cycle 2004–2008.
 - **Placebo period:** election cycle 1990–1994 to 1995–1999.
- **Why use changes.** Election outcomes are observed once per election type per municipality in each election cycle. The empirical design uses first differences between the pre-Internet and Internet periods to remove permanent municipality-level differences in political culture, demographics, and geography.

2.3 Socioeconomic controls

- **Labor-market and education variables.** Social-security records provide wages, education groups, and the foreign-worker share at the municipality level.
- **Demographic variables.** The German Federal Statistical Office provides population size and demographic characteristics.
- **Main controls.** The regressions use changes in:
 - female population share;
 - share of population aged 18–65;
 - share of population aged 65 and older;
 - real daily wage;
 - net migration rate;
 - share of foreigners;
 - unemployment rate;
 - shares of unskilled, skilled, and high-skilled workers.
- **Purpose.** These controls help absorb changes in socioeconomic conditions that might affect both broadband demand and voting behavior.

2.4 Key descriptive facts

- **Turnout decline.** Average voter turnout falls from **73.34%** in 1995–1999 to **64.95%** in 2004–2008, a decline of **8.39 percentage points**.
- **DSL diffusion.** DSL availability rises from zero in the pre-Internet period to an average of about **73% of households** in the Internet period.
- **Party shares.**
 - Established parties fall from **84.79%** to **82.20%**.
 - Small parties rise from **11.76%** to **13.59%**.
 - Right-fringe parties fall from **3.24%** to **2.30%**.
 - Left-fringe parties rise from **0.20%** to **1.91%**.
- **Interpretation.** These raw changes show large movements in political outcomes during the broadband diffusion period, but they do not by themselves identify Internet effects because the locations with broadband may have been changing differently for other reasons.

2.5 Mechanism data

- **Newspapers.** The paper uses municipality-level newspaper circulation from the German Audit Bureau of Circulations for 2000–2008. Newspapers are grouped into local daily, leading national, yellow-press, other daily, and Sunday newspapers.
- **SOEP survey.** The German Socio-Economic Panel provides household and individual information on DSL subscriptions, TV consumption, and time spent on entertainment.
- **Why mechanism data matter.** The election results show whether DSL changes voting behavior. The newspaper and survey data help explain *why*: whether the Internet substitutes for existing media or expands entertainment time.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline first-difference specification

The main estimating equation is

$$\Delta E_{ie} = \alpha + \beta_1 \Delta \text{DSL}_i + \Delta X_i' \beta_2 + \alpha_t + \alpha_e + \varepsilon_{ie}.$$

- i indexes municipalities.
- e indexes election types: federal, state, or local.
- ΔE_{ie} is the change in an election outcome from 1995–1999 to 2004–2008.
- ΔDSL_i is the change in DSL availability. Since DSL availability is zero in the pre-Internet period, this is just DSL availability in the Internet period.

- ΔX_i is the vector of changes in socioeconomic controls.
- α_t are election-year dummies.
- α_e are election-type dummies.

Interpretation. This specification compares how voting outcomes changed in municipalities that received more or less broadband access. First differencing removes permanent municipality characteristics, such as historical political culture or fixed remoteness. The remaining concern is that DSL rollout may still be correlated with time-varying shocks, which motivates the IV design.

3.2 First stage

The first-stage equation is

$$\Delta \text{DSL}_i = \alpha + \gamma_1 \text{PSTN}_i + \Delta X_i' \gamma_2 + \alpha_t + \alpha_e + \nu_{ie}.$$

- PSTN_i denotes a technological feature of the preexisting public switched telephone network.
- The exact definition of PSTN_i differs across the three instruments.
- The expected sign is negative: technological constraints make DSL rollout more difficult.

Interpretation. The first stage asks whether old telephone-network constraints predict lower broadband availability. The identifying variation comes from features that mattered for DSL but were not chosen with Internet demand or later political outcomes in mind.

3.3 Instrument 1: 4,200-meter threshold

For West German municipalities without their own main distribution frame, the first instrument is

$$Z_i^{(1)} = \mathbf{1}\{\text{dist}(i, \text{MDF}_i) > 4200 \text{ meters}\}.$$

- MDF_i is the main distribution frame serving municipality i .
- DSL quality falls sharply once the copper-wire distance from the household to the MDF exceeds about 4,200 meters.
- The sample contains **3,333** less-agglomerated West German municipalities without their own MDF and connected to the closest MDF.
- The regressions include **MDF-by-election-type fixed effects**, so comparisons are between municipalities served by the same MDF catchment area.

Interpretation. This compares “lucky” and “unlucky” municipalities near the same telephone exchange. They share the same local network node, but some happen to lie just too far away for cheap DSL provision.

3.4 Instrument 2: no closer MDF

For West German municipalities whose actual MDF is more than 4,200 meters away, the second instrument is

$$Z_i^{(2)} = \mathbf{1}\{\text{dist}(i, \text{MDF}_i) > 4200 \text{ and no alternative MDF lies within 4200 meters}\}.$$

- Some municipalities are connected to a distant MDF even though another MDF is geographically closer.
- If a closer MDF is available within 4,200 meters, DSL can be provided more easily through reconnection.
- If no closer MDF is available, DSL rollout is more costly.
- The sample contains **1,800** less-agglomerated West German municipalities without their own MDF and with distance to the actual MDF above 4,200 meters.
- The regressions control for distance to the actual MDF and distance to the nearest urban center.

Interpretation. This design isolates variation among municipalities already disadvantaged by being far from their assigned MDF. Some can cheaply escape that constraint by reconnecting to a nearby MDF; others cannot.

3.5 Instrument 3: OPAL areas in East Germany

For East Germany, the instrument is

$$Z_i^{(3)} = \mathbf{1}\{i \text{ is in an original OPAL area}\}.$$

- After reunification, parts of East Germany received OPAL fiber technology.
- OPAL was initially viewed as a state-of-the-art telecommunications technology.
- Later, it became a problem because OPAL was incompatible with DSL.
- The sample contains **1,249** East German municipalities within 4,200 meters of their MDF.
- The regressions include county fixed effects to compare municipalities within the same regional environment.

Interpretation. OPAL areas were unlucky in the DSL era. Their infrastructure looked advanced in the early 1990s but became a bottleneck for mass broadband Internet.

3.6 Local average treatment effects

Because each instrument shifts DSL availability only for a subset of municipalities, the paper interprets its estimates as local average treatment effects:

$$\text{LATE}_j = E[\beta_i \mid \text{municipality } i \text{ is compliant with instrument } j], \quad j = 1, 2, 3.$$

Interpretation. The estimates do not necessarily describe the effect of broadband for every German municipality. They describe the effect for municipalities whose DSL availability was changed by the relevant technological constraint.

3.7 Nonlocal versus local election specification

The paper also splits the sample into nonlocal and local elections:

$$\Delta E_{ie}^g = \alpha^g + \beta_1^g \Delta \text{DSL}_i + \Delta X_i' \beta_2^g + \text{fixed effects} + \varepsilon_{ie}^g, \quad g \in \{\text{nonlocal, local}\}.$$

- Nonlocal elections are federal and state elections.
- Local elections are municipal elections.

Interpretation. This split is important because TV is primarily a source of nonlocal political information in Germany. If the Internet reduces turnout by crowding out TV, the effect should be stronger for nonlocal elections.

3.8 Newspaper mechanism specification

For newspaper circulation, the outcome becomes the change in circulation between 2000 and 2008:

$$\Delta C_i^k = \alpha + \theta_k \Delta \text{DSL}_i + \Delta X_i' \pi_k + \text{fixed effects} + u_i^k,$$

where k indexes newspaper categories.

Interpretation. If the Internet demobilizes voters by replacing newspapers, DSL availability should reduce newspaper circulation. The paper finds little evidence for this channel.

3.9 Survey mechanism specification

Using the SOEP household survey, the paper studies actual DSL subscriptions and media use:

$$Y_h = \alpha + \theta \text{DSLSub}_h + W_h' \pi + \delta_c + \eta_h,$$

with first stage

$$\text{DSLSub}_h = \alpha + \delta_1 Z_h^{(1)} + \delta_2 Z_h^{(2)} + \delta_3 Z_h^{(3)} + W_h' \rho + \delta_c + \xi_h.$$

- h indexes households or individuals.
- Y_h is TV consumption or time spent on entertainment.
- DSLSub_h is an indicator for having a DSL subscription.
- W_h includes individual and household controls.
- δ_c are county fixed effects.

Interpretation. This specification checks whether the technical availability instruments predict actual household DSL subscriptions, and whether DSL subscriptions change media consumption. The estimates support the mechanism that DSL crowds out TV and raises entertainment time.

3.10 Turnout magnitude calculation

The paper interprets the turnout effect by decomposing turnout in the Internet period:

$$T_1 = \gamma T_1^I + (1 - \gamma) T_1^{NI},$$

where

- T_1 is observed turnout in the Internet period;
- T_1^I is turnout among people exposed to broadband Internet;
- T_1^{NI} is turnout among people not exposed to broadband Internet;
- γ is the population share exposed to broadband, approximated by DSL availability.

The no-Internet counterfactual turnout for the nonexposed group is

$$\hat{T}_1^{NI} = T_1 - \hat{\beta}_1 \text{DSL}.$$

Solving for turnout among exposed individuals gives

$$\hat{T}_1^I = \frac{T_1 - (1 - \text{DSL})\hat{T}_1^{NI}}{\text{DSL}}.$$

Interpretation. The average municipal turnout effect understates the effect on people who actually receive Internet access. The paper's calculation implies a demobilization effect of about **8.3 percentage points** in the LATE 1 sample and **8.1 percentage points** in the LATE 2 sample.

4 Identification and instruments (what makes the estimates credible)

4.1 What would fail in a simple approach

- Municipalities with high broadband availability are not random.
- Telecom firms prefer to roll out broadband where demand is high.
- High demand is likely in richer, younger, more educated municipalities.
- These same characteristics are correlated with voter turnout and party support.
- Therefore, a simple regression of turnout on DSL availability would likely be biased upward for turnout.

Key point. The paper's OLS estimates are indeed much closer to zero than the IV estimates, consistent with the idea that broadband was disproportionately available in municipalities where turnout would otherwise have been higher.

4.2 Why the telephone-network instruments are plausible

- **Timing.** The West German voice-telephony network was largely laid out in the 1960s, long before broadband Internet existed.
- **Purpose.** The network was designed for voice telephony under a state monopoly with a universal-service goal, not for Internet access or future political behavior.
- **Distance irrelevance for voice.** Copper-wire distance mattered little for ordinary telephone service, but it mattered sharply for DSL signal quality.
- **Institutional detail.** Locations of main distribution frames in less-agglomerated areas were influenced by lot and building availability, cable ducts, and administrative constraints.
- **Within-network comparisons.** The LATE 1 design compares municipalities served by the same MDF, using MDF-by-election-type fixed effects.
- **Alternative-distance robustness.** The results are robust to measuring distance from the geographic centroid or from the population center.
- **Narrow-band robustness.** The LATE 1 result remains when the sample is restricted to municipalities within 2,000 meters of the 4,200-meter threshold.

4.3 Why the OPAL instrument is plausible

- OPAL was rolled out in East Germany after reunification as part of infrastructure modernization.
- It was viewed at the time as advanced technology, not as a future broadband disadvantage.
- The incompatibility with DSL was an ex post technological accident.
- The East German regressions include county fixed effects to absorb regional differences.

Caveat. The East German estimates are much less precise. East Germany had a shorter post-socialist democratic history in the pre-Internet period, more volatile voting behavior, and less established media-consumption patterns.

4.4 First-stage strength

- **LATE 1:** being above the 4,200-meter threshold lowers DSL availability by about **10.3–15.0 percentage points**; first-stage F -statistics are about **140–263**.
- **LATE 2:** having no closer MDF lowers DSL availability by about **13.4–13.9 percentage points**; first-stage F -statistics are about **150–192**.
- **LATE 3:** being in an OPAL area lowers DSL availability by about **7.9–8.8 percentage points**; first-stage F -statistics are about **9–12**.

Interpretation. The West German instruments are very strong. The East German OPAL instrument is relevant but weaker, which partly explains the imprecision of the East German estimates.

4.5 Validity checks

- **Placebo election trends.** The instruments do not predict changes in turnout or party vote shares between 1990–1994 and 1995–1999, before broadband Internet existed.
- **Socioeconomic outcomes.** The instruments generally do not predict changes in socioeconomic controls during the broadband period.
- **Controls.** Adding changes in demographic and labor-market controls has little effect on the IV coefficients.
- **Alternative samples.** The results are robust across two independent West German instruments.
- **Mechanism consistency.** The negative turnout effects are concentrated in the places and elections where the media-crowding-out mechanism is most plausible: West Germany and nonlocal elections.

4.6 Relation to the literature

- The paper contributes to the literature on media and politics by studying the Internet as a new mass medium.
- It parallels studies of radio, newspapers, and television that use technological or geographic variation in media access.
- It is especially close to research showing that television can lower voter turnout by crowding out more informative media.
- It also relates to the literature on ideological segregation in media consumption: if Internet users seek like-minded sources, more information does not necessarily shift vote shares.

5 Tables and figures

5.1 Figure 1: development of broadband Internet

Read:

- The figure shows histograms of DSL availability across German municipalities from 2005 to 2008.
- The distribution shifts upward over time, showing rapid diffusion of broadband Internet.
- There remains substantial cross-municipality variation in each year.
- This variation is the raw input for the empirical design, but the paper uses instruments to isolate the part caused by technological constraints rather than demand.

5.2 Table 1: descriptive statistics

Read:

- Turnout falls by **8.39 percentage points** between the pre-Internet and Internet periods.

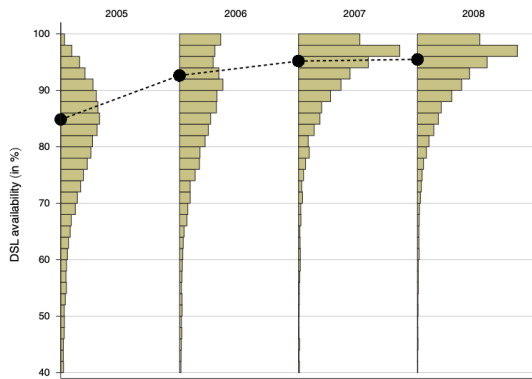


FIGURE 1. DEVELOPMENT OF BROADBAND INTERNET

Notes: Chart shows histograms of DSL availability (measured as a percentage of households for which DSL is technically available) in all German municipalities for the different years. The chart is truncated at 40 percent. The dotted line connects the population-weighted mean availabilities for all years.

TABLE 1—DESCRIPTIVE STATISTICS

	Pre-Internet (1995–1999)		Internet (2004–2008)		Change	
	Mean (1)	SD (2)	Mean (3)	SD (4)	Mean (3) – (1) (5)	SD (6)
<i>Outcome variables</i>						
Voter turnout	73.34	9.30	64.95	11.27	–8.39	7.61
Vote share established parties	84.79	21.37	82.20	22.17	–2.59	8.94
Vote share small parties	11.76	22.57	13.59	23.58	1.83	8.28
Vote share right-fringe parties	3.24	3.42	2.30	2.95	–0.94	2.48
Vote share left-fringe parties	0.20	0.46	1.91	2.51	1.71	2.33
<i>Broadband Internet</i>						
DSL availability (share of households)	0.00	0.00	0.73	0.29	0.73	0.29
<i>Control variables</i>						
Female population share	50.20	1.80	50.10	1.82	–0.10	1.00
Share of population aged 18–65	67.03	3.37	64.38	5.75	–2.65	4.64
Share of population aged 65 and older	16.61	3.33	19.11	3.62	2.50	1.71
Real daily wage (in 2008 €)	88.49	13.15	84.70	12.89	–3.79	3.21
Share of net migration	0.21	1.97	–0.25	1.77	–0.46	2.52
Share of foreigners	2.77	3.86	2.74	3.83	–0.03	1.45
Unemployment rate	6.73	4.55	7.63	5.06	0.90	2.15
Share of unskilled workers	11.17	6.14	9.34	5.28	–1.83	2.83
Share of skilled workers	81.99	6.23	82.68	6.10	0.69	3.21
Share of high-skilled workers	6.84	4.13	7.97	4.61	1.13	2.04
Distance to next urban center (in km)	26.04	15.35	26.04	15.35	0.00	0.00
Municipalities	11,832		11,832		11,832	
Observations	35,391		35,391		35,391	

Notes: Table reports descriptive statistics for the full sample of German municipalities across three types of elections (federal, state, local). Values in percent, unless otherwise specified. The number of observations does not add up to three times the number of municipalities because we have 103 missing observations for local elections in Baden-Wuerttemberg and 2 missing observations for local elections in the city-states of Berlin and Hamburg where local and state elections coincide. Columns 1 and 2 report means and standard deviations from the period before broadband Internet introduction (1995–1999) and columns 3 and 4 report the analogous values for the period after the introduction of broadband Internet (2004–2008). DSL availability refers to the years 2005–2008. Control variables in the pre-Internet period refer to the years 2000 (foreigners, skills, unemployment, wages); or 2001 (remaining variables). Finally, columns 5 and 6 report changes between the two periods.

- DSL availability rises from **0** to **0.73** on average.
- Established parties lose vote share, small parties gain vote share, right-fringe parties decline, and left-fringe parties rise.
- The descriptive pattern motivates the question, but it cannot identify causality because many other things changed over the same period.

5.3 Figure 2: graphical illustration of the distance instruments

Read:

- Panel A illustrates the 4,200-meter threshold around an MDF.
- Municipalities served by the same MDF can differ sharply in DSL feasibility depending on whether they lie inside or outside the threshold.
- Panel B illustrates the “no closer MDF” design.
- Two municipalities may both be far from their assigned MDF, but only one may have access to a closer alternative MDF.

5.4 Table 2: basic results for the LATE 1 sample

Read:

- The LATE 1 sample uses West German municipalities without an own MDF.
- The instrument is the 4,200-meter threshold.
- First-stage coefficients are strongly negative: being above the threshold lowers DSL availability by roughly **10–15 percentage points**.
- IV estimates show a robust negative effect on turnout:

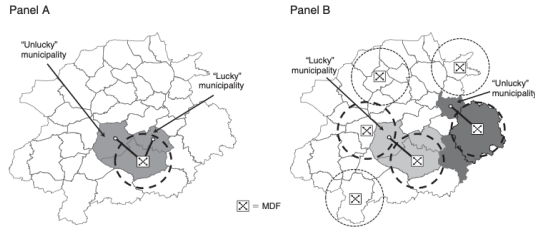


FIGURE 2. GRAPHICAL ILLUSTRATION OF THE DISTANCE INSTRUMENTS

Notes: In panel A, the three gray-shaded West German municipalities are served by the same MDF. The circle represents the 4,200 m threshold around the MDF. While large parts of the northwestern municipality without an own MDF lies above the threshold, large parts of the northeastern municipality without an own MDF lies below the threshold. As a result, technical DSL availability is higher in the northeastern municipality than in the northwestern municipality. In panel B, the map illustrates the catchment areas (light gray and dark gray shaded) of two different West German MDFs. The circles represent the 4,200 m threshold around an MDF. In both catchment areas, large parts of the northwestern municipalities lie above the threshold. However, while for the light gray shaded northwestern municipality a significantly closer MDF is available, the actual MDF is also the closest MDF for the dark gray shaded northwestern municipality. As a result, technical DSL availability is higher in the light gray shaded northwestern municipality that can be reconnected to a closer MDF than in the dark gray shaded northwestern municipality lacking this option.

TABLE 2—BASIC ESTIMATION RESULTS FOR THE LATE 1 SAMPLE

	OLS First-difference model		IV Geographic centroid		IV 2,000m around threshold		IV Population center	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Δ voter turnout	-0.401 (0.334)	-0.407 (0.332)	-3.347*** (0.984)	-3.558*** (0.989)	-4.380*** (1.488)	-4.772*** (1.501)	-3.219*** (0.949)	-3.423*** (0.953)
Δ vote share established parties	-0.304 (0.285)	-0.329 (0.284)	-1.257 (0.971)	-1.368 (0.973)	-0.283 (1.448)	-0.440 (1.456)	-1.054 (0.930)	-1.081 (0.929)
Δ vote share small parties	0.333 (0.234)	0.342 (0.234)	1.962** (0.868)	2.009** (0.870)	1.617 (1.292)	1.630 (1.300)	1.682** (0.821)	1.658** (0.821)
Δ vote share right-fringe parties	0.045 (0.107)	0.055 (0.106)	-0.312 (0.263)	-0.277 (0.268)	-0.477 (0.394)	-0.369 (0.402)	-0.063 (0.259)	-0.033 (0.263)
Δ vote share left-fringe parties	-0.073 (0.106)	-0.067 (0.106)	-0.392 (0.304)	-0.364 (0.308)	-0.857* (0.462)	-0.820* (0.465)	-0.565* (0.299)	-0.544* (0.301)
First stage Threshold	—	—	-0.142*** (0.009)	-0.141*** (0.009)	-0.104*** (0.009)	-0.103*** (0.009)	-0.150*** (0.009)	-0.149*** (0.009)
F-test of excluded instruments	—	—	255.11	252.88	141.98	140.22	262.86	262.22
Control variables	No	Yes	No	Yes	No	Yes	No	Yes
Number of MDFs	869	869	869	869	742	742	869	869
Municipalities	3,333	3,333	3,333	3,333	2,649	2,649	3,339	3,339
Observations	9,923	9,923	9,923	9,923	7,882	7,882	9,941	9,941

Notes: Table reports municipality-level pooled regressions for elections at three main levels of governance in Germany: federal elections, state elections, and local elections. The sample consists of West German municipalities without an own MDF which are actually connected to the closest MDF. Each cell in the upper part of the table shows the coefficient from a separate regression of an election outcome on DSL availability. Columns 1 and 2 show first-difference estimates; columns 3 and 4 show instrumental-variable estimates where distance calculations are based on municipalities' geographic centroid; and columns 5 and 6 show the same regressions on a sample that is restricted to a band of 2,000 m around the threshold of 4,200 m, i.e., between 2,200 and 6,200 m from the MDF. Columns 7 and 8 show instrumental-variable estimates with an alternative distance measure based on municipalities' population center. Since we exclude municipalities that could be connected to a closer MDF from the LATE 1 sample, the sample size changes when we employ another distance measure. The instrument in columns 3–8 is a threshold dummy indicating whether a municipality is more than 4,200m away from its MDF (1 = lower probability of DSL availability), and 0 otherwise. The F-test of excluded instruments refers to the Kleibergen-Paap F-statistic. The five election outcomes are the changes in voter turnout, vote share of established parties, vote share of small parties, vote share of right-fringe parties, and vote share of left-fringe parties between the pre-Internet period (1995–1999) and the Internet period (2004–2008). All specifications include election type dummies, MDF-by-election-type dummies, and year dummies. Specifications in columns 2, 4, 6, and 8 additionally control for changes in the following municipality characteristics: female population share, share of population aged 18–65, share of population older than 65, average wage, net migration rate, share of foreigners, unemployment rate, and share of unskilled, skilled, and high-skilled workers. Descriptive statistics for the LATE 1 sample are provided in online Appendix Table F2. All standard errors are clustered on the municipality level.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

- about –3.35 to –3.56 percentage points in the full geographic-centroid specification;
- about –4.38 to –4.77 percentage points in the narrow threshold band;
- about –3.22 to –3.42 percentage points using the population-center distance measure.

- The OLS estimates are small and insignificant, consistent with upward selection bias.
- Party vote-share results are mixed; the paper does not interpret them as robust evidence that DSL benefits a particular party family.

5.5 Table 3: basic results for the LATE 2 sample

Read:

- The LATE 2 sample focuses on West German municipalities more than 4,200 meters from their assigned MDF.
- The instrument is whether the municipality lacks a closer alternative MDF.
- First-stage coefficients are again strong: having no closer MDF lowers DSL availability by about **13–14 percentage points**.
- The turnout effects are very similar to LATE 1:
 - about –3.68 and –3.37 percentage points using geographic-centroid distance;
 - about –2.85 and –3.10 percentage points using population-center distance.
- The similarity between LATE 1 and LATE 2 is important because the two West German instruments use different sources of technical variation.

- Party vote-share coefficients are not systematically significant.

TABLE 3—BASIC ESTIMATION RESULTS FOR THE LATE 2 SAMPLE

	OLS		IV		IV	
	First-difference model		Geographic centroid		Population center	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ voter turnout	0.525 (0.352)	0.387 (0.366)	−3.684** (1.795)	−3.367* (1.835)	−2.845* (1.570)	−3.096* (1.640)
Δ vote share established parties	0.755* (0.402)	0.741* (0.420)	−1.805 (2.286)	−2.099 (2.293)	−0.267 (2.021)	−0.286 (2.069)
Δ vote share small parties	0.111 (0.346)	0.053 (0.360)	1.142 (2.037)	1.468 (2.045)	1.209 (1.835)	1.224 (1.878)
Δ vote share right-fringe parties	−0.370*** (0.125)	−0.447*** (0.130)	−0.075 (0.687)	0.124 (0.689)	−0.734 (0.580)	−0.750 (0.597)
Δ vote share left-fringe parties	−0.497*** (0.130)	−0.347*** (0.132)	0.738 (0.555)	0.508 (0.540)	−0.208 (0.509)	−0.189 (0.535)
<i>First stage</i>						
“No Closer MDF”	—	—	−0.138*** (0.010)	−0.136*** (0.010)	−0.139*** (0.011)	−0.134*** (0.011)
<i>F</i> -test of excluded instruments	—	—	191.98	180.40	163.31	150.38
Control variables	No	Yes	No	Yes	No	Yes
Municipalities	1,800	1,800	1,800	1,800	1,742	1,742
Observations	5,362	5,362	5,362	5,362	5,191	5,191

Notes: Table reports municipality-level pooled regressions for elections at three main levels of governance in Germany: federal elections, state elections, and local elections. The sample consists of West German municipalities without an own MDF which are connected to an MDF above the 4,200 m threshold. Each cell in the upper part of the table shows the coefficient from a separate regression of an election outcome on DSL availability. Columns refer to different models and rows refer to different outcome variables and test statistics. Columns 1 and 2 show first-difference estimates; columns 3 and 4 show instrumental-variable estimates where distance calculations are based on municipalities' geographic centroid; and columns 5 and 6 show instrumental-variable estimates with an alternative distance measure based on municipalities' population center. The instrument in columns 3–6 is a dummy variable that equals unity if a municipality is connected to an MDF at more than 4,200 m distance and cannot be connected to another MDF that is closer than 4,200 m (1 = lower probability of DSL availability), and zero otherwise. The *F*-test of excluded instruments refers to the Kleibergen-Paap *F*-statistic. The five election outcomes are the changes in voter turnout, vote share for established parties, vote share for small parties, vote share for right-fringe parties, and vote share for left-fringe parties between the pre-Internet period (1995–1999) and the Internet period (2004–2008). All specifications include election type dummies, year dummies, and controls for the distance to the actual MDF and the distance to the next urban center. Specifications in columns 2, 4, and 6 additionally control for changes in the following municipality characteristics: female population share, share of population aged 18–65, share of population older than 65, average wage, net migration rate, share of foreigners, unemployment rate, and share of unskilled, skilled, and high-skilled workers. Descriptive statistics for the LATE 2 sample are provided in the online Appendix's Table F3. All standard errors are clustered on the municipality level.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

TABLE 4—BASIC ESTIMATION RESULTS FOR THE LATE 3 SAMPLE

	OLS		IV		IV	
	First-difference model		Geographic centroid		Population center	
	(1)	(2)	(3)	(4)	(5)	(6)
Δ voter turnout	0.180 (0.395)	0.153 (0.394)	1.696 (4.261)	0.730 (4.172)	−0.160 (3.774)	−0.922 (3.784)
Δ vote share established parties	1.230* (0.671)	1.227* (0.672)	1.242 (7.621)	0.850 (7.513)	−0.350 (6.597)	−0.607 (6.607)
Δ vote share small parties	−0.951 (0.660)	−0.967 (0.659)	0.128 (7.440)	0.515 (7.318)	2.105 (6.471)	2.410 (6.466)
Δ vote share right-fringe parties	−0.277** (0.133)	−0.261* (0.134)	−1.071 (1.673)	−1.076 (1.613)	−1.400 (1.498)	−1.456 (1.485)
Δ vote share left-fringe parties	−0.002 (0.016)	0.000 (0.016)	−0.299 (0.195)	−0.289 (0.191)	−0.355* (0.182)	−0.348* (0.182)
<i>First stage</i>						
OPAL	—	—	−0.079*** (0.026)	−0.080*** (0.026)	−0.088*** (0.026)	−0.088*** (0.026)
<i>F</i> -test of excluded instruments	—	—	9.05	9.54	11.50	11.61
Control variables	No	Yes	No	Yes	No	Yes
Number of counties	63	63	63	63	63	63
Municipalities	1,249	1,249	1,249	1,249	1,279	1,279
Observations	3,747	3,747	3,747	3,747	3,837	3,837

Notes: Table reports municipality-level pooled regressions for elections at three main levels of governance in Germany: federal elections, state elections, and local elections. The sample consists of East German municipalities with distance to the actual MDF below 4,200 m. Each cell in the upper part of the table shows the coefficient from a separate regression of an election outcome on DSL availability. Columns refer to different models and rows refer to different outcome variables and test statistics. Columns 1 and 2 show first-difference estimates; columns 3 and 4 show instrumental-variable estimates where distance calculations are based on municipalities' geographic centroid; and columns 5 and 6 show instrumental-variable estimates with an alternative distance measure based on municipalities' population center. The instrument in columns 3–6 is a dummy variable that equals unity if a municipality was initially supplied with OPAL technology (1 = lower probability of DSL availability), and zero otherwise. The *F*-test of excluded instruments refers to the Kleibergen-Paap *F*-statistic. The five election outcomes are the changes in voter turnout, vote share of established parties, vote share of small parties, vote share of right-fringe parties, and vote share of left-fringe parties between the pre-Internet period (1995–1999) and the Internet period (2004–2008). All specifications include election type dummies, county-by-election-type dummies, and year dummies. Specifications in columns 2, 4, and 6 additionally control for changes in the following municipality characteristics: female population share, share of population aged 18–65, share of population older than 65, average wage, net migration rate, share of foreigners, unemployment rate, and share of unskilled, skilled, and high-skilled workers. Descriptive statistics for the LATE 3 sample are provided in online Appendix Table F4. All standard errors are clustered on the municipality level.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

5.6 Table 4: basic results for the LATE 3 sample

Read:

- The LATE 3 sample studies East German municipalities and uses OPAL areas as the instrument.
- OPAL lowers DSL availability by about **8 percentage points**.
- The first stage is weaker than in the West German designs.
- The turnout coefficient is not statistically significant.
- The party-share coefficients are also generally insignificant.
- The paper interprets this as evidence that the negative turnout result is a West German result, possibly because East German media and political behavior were still shaped by the post-socialist transition.

5.7 Table 5: validity tests

Read:

- Panel A runs placebo regressions for pre-Internet election changes.
- The instruments do not predict turnout or vote-share changes before broadband existed.
- Panel B checks whether instruments predict changes in socioeconomic variables during the broadband period.

- Most coefficients are insignificant, suggesting that the instruments are not proxying for broad local economic changes.
- This supports the exclusion restriction.

TABLE 5—VALIDITY TESTS

	LATE 1 Threshold (1)	LATE 2 No closer MDF (2)	LATE 3 OPAL (3)
<i>Panel A. Pre-Internet period</i>			
Δ voter turnout	-0.105 (0.113)	-0.012 (0.167)	0.081 (0.360)
Δ vote share established parties	-0.060 (0.114)	0.195 (0.197)	0.011 (0.662)
Δ vote share small parties	0.085 (0.100)	-0.134 (0.182)	-0.208 (0.643)
Δ vote share right-fringe parties	-0.043 (0.036)	-0.007 (0.077)	0.180 (0.144)
Δ vote share left-fringe parties	0.001 (0.010)	-0.009 (0.016)	0.003 (0.004)
<i>Panel B. Broadband-Internet period</i>			
Δ female population share	0.002 (0.046)	-0.052 (0.073)	-0.137 (0.088)
Δ share of population aged 18–65	0.178** (0.070)	0.117 (0.100)	-0.137 (0.121)
Δ share of population aged 65 and older	-0.058 (0.052)	-0.052 (0.073)	0.027 (0.097)
Δ daily wage (2008 = 1)	-0.193 (0.122)	-0.103 (0.215)	0.029 (0.214)
Δ share of net migration	0.042 (0.112)	0.143 (0.150)	-0.098 (0.214)
Δ share of foreigners	0.056 (0.070)	0.199* (0.106)	0.071 (0.045)
Δ unemployment rate	-0.039 (0.053)	0.010 (0.080)	0.330 (0.230)
Δ share of unskilled workers	0.012 (0.131)	-0.106 (0.183)	0.155 (0.201)
Δ share of skilled workers	0.085 (0.150)	0.299 (0.208)	-0.141 (0.251)
Δ share of high skilled workers	-0.097 (0.085)	-0.193* (0.115)	-0.014 (0.161)
Municipalities	3,333	1,800	1,249
Observations	9,923	5,362	3,747

Notes: Table reports municipality-level pooled regressions for two types of validity tests. Values in percent. Panel A reports reduced-form placebo regressions of changes in election outcomes in the pre-Internet period, i.e., between the period 1990–1994 and 1995–1999, on the three instruments. Each cell shows the coefficient from a separate regression. Columns refer to different models and rows refer to different outcome variables. In LATE 1, the variable of interest is the threshold dummy at 4,200 m; in LATE 2, the variable of interest is the “No Closer MDF” dummy; and in LATE 3, the variable of interest is the OPAL dummy. All estimations control for changes in the municipalities’ population and share of females along with election type dummies, and year dummies. Column 1 estimations additionally include MDF-by-election-type dummies; column 2 estimations include controls for the distance to the actual MDF and the distance to the next urban center; and column 3 estimations include county dummies. Panel B reports reduced-form estimations of the first differences of municipalities’ socioeconomic characteristics on the three instruments. The differences are calculated between the pre-Internet period (1995–1999) and the Internet period (2004–2008). All regressions include the same controls as those in the basic regressions except the one that is the dependent variable. In the skill-level regressions, we do not include the shares of unskilled, skilled, and high-skilled workers as controls, since they add up to one. Standard errors are clustered on the municipality level.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

TABLE 6—ESTIMATION RESULTS FOR SUBSAMPLES

	LATE 1 sample		LATE 2 sample		LATE 3 sample	
	Nonlocal (1)	Local (2)	Nonlocal (3)	Local (4)	Nonlocal (5)	Local (6)
Δ voter turnout	-4.262*** (0.991)	-1.999 (1.753)	-5.026** (2.059)	-0.817 (2.774)	-1.962 (3.578)	10.208 (13.572)
Δ vote share established parties	0.577 (0.690)	-5.529** (2.589)	-1.447 (1.603)	-4.582 (6.210)	-1.023 (2.498)	6.673 (33.282)
Δ vote share small parties	0.339 (0.405)	5.583** (2.574)	0.753 (0.855)	4.385 (6.175)	2.816* (1.646)	-7.017 (33.373)
Δ vote share right-fringe parties	-0.600 (0.382)	0.410** (0.173)	0.095 (0.930)	0.052 (0.625)	-1.436 (1.886)	0.344 (1.905)
Δ vote share left-fringe parties	-0.316 (0.422)	-0.464* (0.280)	0.599 (0.749)	0.144 (0.591)	-0.357 (0.236)	—
<i>First stage</i> Instrument	-0.143*** (0.009)	-0.137*** (0.009)	-0.137*** (0.010)	-0.129*** (0.011)	-0.095*** (0.027)	-0.052* (0.028)
<i>F</i> -test of excluded instruments	220.82	249.88	180.43	129.03	12.71	3.53
Number of MDFs/counties	869	853	—	—	63	63
Municipalities	3,333	3,257	1,800	1,762	1,249	1,249
Observations	6,666	3,257	3,600	1,762	2,498	1,249

Notes: Table reports municipality-level instrumental-variable regressions for nonlocal (federal and state) and local elections. Each cell in the upper part of the table shows the coefficient from a separate regression of an election outcome on DSL availability. Columns refer to different models and rows refer to different outcome variables and test statistics. Columns 1 and 2 show estimates for the LATE 1 sample with the threshold at 4,200 m as instrument; columns 3 and 4 show estimates for the LATE 2 sample with “no closer MDF” as instrument; and columns 5 and 6 show estimates for the LATE 3 sample with OPAL as instrument. The control variables and sample definitions are detailed in the notes to Tables 2–4. The *F*-test of excluded instruments refers to the Kleibergen-Paap *F*-statistic. Standard errors are clustered on the municipality level in columns 1, 3, and 5, and robust in columns 2, 4, and 6, since we observe only one local election per municipality.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

5.8 Table 6: nonlocal versus local elections

Read:

- The negative turnout effect is concentrated in nonlocal elections.
- In the LATE 1 sample, DSL lowers turnout by about **4.26 percentage points** in nonlocal elections, but the local-election effect is smaller and insignificant.
- In the LATE 2 sample, DSL lowers turnout by about **5.03 percentage points** in nonlocal elections, but again not significantly in local elections.
- This pattern fits the mechanism: TV is more important for federal and state political information than for local political information.

5.9 Table 7: newspapers

Read:

- The table estimates effects of DSL on newspaper circulation between 2000 and 2008.
- There is no strong evidence that broadband reduces circulation of daily newspapers.
- The only notable significant coefficient is a positive effect on yellow-press newspapers in the LATE 1 sample.
- Therefore, the paper does not attribute the turnout decline to a reduction in newspaper subscriptions.
- A caveat is that circulation data cannot reveal whether people spend less time reading newspapers they still receive.

	LATE 1 Threshold (1)	LATE 2 No closer MDF (2)	LATE 3 OPAL (3)
<i>Difference 2000–2008</i>			
Δ All daily newspapers	1.648 (3.398)	2.805 (15.288)	−0.812 (2.959)
Δ Local daily newspapers	−2.917 (1.987)	0.269 (5.919)	2.440 (2.314)
Δ Other daily newspapers	0.843 (1.753)	−3.920 (12.740)	−1.534 (1.769)
Δ National daily newspapers	0.398 (0.818)	0.990 (0.602)	0.242 (0.234)
Δ Yellow press daily newspapers	3.243** (1.588)	5.970 (4.564)	−1.961 (1.617)
Δ Sunday newspapers	−0.045 (0.187)	1.281 (1.588)	0.115* (0.064)
<i>First stage</i>			
Instrument	−0.129*** (0.009)	−0.110*** (0.009)	−0.242*** (0.030)
<i>F</i> -test of excluded instruments	197.04	145.55	64.26
Number of observations/municipalities	3,319	1,792	1,247

Notes: Table reports municipality-level instrumental-variable regressions for changes in the newspaper circulation between 2000 and 2008. Each cell in the upper part of the table shows the coefficient from a separate regression of a newspaper type on DSL availability. The outcome variables are percentage point changes in the circulation of all daily newspapers, local daily newspapers, other nonlocal newspapers, seven leading national newspapers, yellow-press daily newspapers, and Sunday newspapers. All outcomes are standardized by the number of households in a municipality. Column 1 refers to instrumental-variable estimations for the LATE 1 sample with the threshold at 4,200 m as instrument; column 2 shows instrumental-variable results for the LATE 2 sample with “no closer MDF” as instrument; and column 3 shows instrumental-variable results for the LATE 3 sample with OPAL as instrument. The *F*-test of excluded instruments refers to the Kleibergen-Paap *F*-statistic. All estimations control for changes in the following municipality characteristics: female population share, share of population aged 18–65, share of population older than 65; average wage, net migration rate, share of foreigners, unemployment rate, and share of unskilled, skilled, and high-skilled workers. LATE 1 estimations additionally include MDF dummies; LATE 2 estimations control for distance to the actual MDF and distance to the next urban center; and LATE 3 estimations include county dummies. Standard errors are robust.

*** Significant at the 1 percent level.
 ** Significant at the 5 percent level.
 * Significant at the 10 percent level.

	TV watching (standardized) (1)	Entertainment (hours)		
		Workday (2)	Saturday (3)	Sunday (4)
DSL subscription	−0.340* (0.212)	0.585 (0.422)	0.858 (0.564)	1.303* (0.727)
<i>First stage</i>				
Threshold	−0.136*** (0.024)	−0.141*** (0.024)	−0.143*** (0.025)	−0.142*** (0.025)
“No closer MDF”	−0.095* (0.048)	−0.109** (0.049)	−0.114** (−0.05)	−0.110** (0.049)
OPAL	−0.124*** (0.029)	−0.125*** (0.029)	−0.125*** (0.029)	−0.128*** (0.029)
<i>F</i> -test of excluded instruments	17.91	17.77	17.92	17.97
Overidentification test	2.004	1.163	0.057	0.594
χ^2 <i>p</i> -value	0.367	0.559	0.972	0.743
Observations	16,043	14,932	14,858	14,851

Notes: Table reports individual-level instrumental-variable regressions with the threshold at 4,200 m, “No closer MDF” and OPAL as instruments. Dependent variable reported on top of each column. Sample includes all adults in SOEP (in 2008) with available information on the respective outcome variable. Additional controls included in each model: the individual’s gender, age (and its square), marital status, number of children living in the household, secondary and university education, occupational status, migration background, a dummy indicating house/flat ownership, and log of net household income. All regressions further include county dummies. The *F*-test for excluded instruments refers to the Kleibergen-Paap *F*-statistic. Standard errors are clustered on the household level. The overidentification test is based on the Huber-White robust variance-covariance matrix without clustering.

*** Significant at the 1 percent level.
 ** Significant at the 5 percent level.
 * Significant at the 10 percent level.

5.10 Table 8: survey evidence on TV and entertainment

Read:

- The paper uses SOEP household data and instruments actual DSL subscriptions with the same network-based instruments.
- The first stage is strong enough at the household level, confirming that technical availability predicts actual DSL subscriptions.
- DSL subscription reduces TV watching by about **0.34 standard deviations**.

- DSL increases entertainment time especially on Sundays: about **1.3 hours**, significant at the 10 percent level.
- The overidentification tests do not reject the joint validity of the instruments.
- This is the main mechanism evidence: the Internet appears to crowd out TV and increase entertainment consumption.

6 Short summary

- **Method.** Compare changes in election outcomes before and after broadband diffusion across German municipalities.
- **Identification.** Use predetermined technological constraints in the old telephone network as instruments for DSL availability.
- **Application.** German federal, state, and local elections, with broadband availability measured at the municipality level from 2005 to 2008.
- **Main turnout result.** DSL availability lowers voter turnout in West Germany by about **3–5 percentage points**, especially in nonlocal elections.
- **Main party-share result.** There is no robust evidence that broadband systematically benefits particular parties or ideological blocs.
- **Mechanism.** The turnout effect is consistent with the Internet crowding out TV consumption and increasing entertainment time, rather than with a large decline in newspaper circulation.
- **Overall takeaway.** Lower information costs do not automatically raise participation. A new medium can demobilize voters when it reallocates attention away from politically informative media and toward entertainment.

7 Conclusion

7.1 Core conclusion

- Broadband Internet availability reduced voter turnout in West Germany during the early broadband period.
- The effect was concentrated in federal and state elections, not local elections.
- Broadband did not systematically shift support toward established parties, fringe parties, or a clear ideological side.
- The paper's IV strategy makes a causal interpretation plausible by relying on technological constraints inherited from the pre-Internet telephone network.

7.2 Economic intuition

- The Internet has two opposing effects.

- On one side, it lowers the cost of accessing political information.
- On the other side, it expands entertainment options and may crowd out incumbent media that expose people to politics incidentally.
- In this setting, the second force dominates for turnout.
- The absence of systematic party effects is consistent with ideological self-selection: people may use the Internet to consume information that matches their prior views, so vote shares do not shift much even if information access expands.

7.3 Takeaway

This paper shows that the political effects of new media depend not only on information availability, but also on attention, substitution across media, and entertainment. In early broadband Germany, the Internet appears to have reduced turnout by crowding out TV and increasing entertainment consumption, while leaving partisan support largely unchanged. The broader lesson is that more information technology does not necessarily mean more political participation.